# Abheejan Lal Shrestha

**Backend Developer**

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## SUMMARY

Backend developer building Go services for Flutter apps, with MongoDB as my preferred database. Computer Engineering student at Pokhara University (NCIT, 2023 to 2027) and first author of a published paper on hybrid search and fraud moderation for rental marketplaces.

## TECHNICAL SKILLS

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| **Backend** | Go, chi, REST APIs, WebSockets, JWT, OIDC, Firebase Admin SDK |
| **Databases** | MongoDB, PostgreSQL, MySQL, sqlc, goose migrations |
| **Mobile and web** | Dart, Flutter, Riverpod, Dio, go_router, React |
| **DevOps** | Docker, Docker Compose, GitHub Actions, Render, Firebase Cloud Messaging |
| **Other languages** | Python (FastAPI), JavaScript, Node.js, Java, C, C++ |

## PROJECTS

**RoomieKTM: room-finding marketplace for Kathmandu** GitHub

Go, MongoDB, Flutter, Firebase, FastAPI, Docker

- Built a Go REST API on chi with MongoDB, verifying Firebase ID tokens on protected requests and sending FCM push notifications.
- Implemented hybrid search that merges BM25 and vector k-NN results with Reciprocal Rank Fusion, then applies fuzzy re-ranking.
- Designed a five-factor trust score that routes listings through automated moderation, flagging 13 of 15 constructed fraud patterns in a pilot.
- Added landlord KYC review, tenant reporting and an append-only audit log, plus a Flutter mobile app and admin panel.

**Revah Management System: collaborative workspace**

Go, PostgreSQL, Flutter Web, Vikunja, Keycloak, Docker

- Built a Go backend-for-frontend on chi that combines its own data with the Vikunja task engine for Flutter Web clients.
- Used pgx and sqlc for type-safe PostgreSQL access, with goose migrations applied at startup.
- Delivered projects, tasks, approvals, budgets, calendars and dashboards across a wide API under /api/v1.
- Added realtime chat over WebSockets, LiveKit calls and Keycloak OIDC sign-in, with Render deployment configuration.

**Academic Portfolio Platform** GitHub

Go, chi, Flutter Web, Riverpod, Docker

- Built a professor's portfolio site with a rate-limited Go API and a Flutter Web frontend that still renders when the API is offline.

**Modern Chess Web App** Live demo

React, Firebase

- Built online play against a bot or friends, with player profiles, achievements and statistics.

## PUBLICATION

**A. L. Shrestha**, P. Mishra, S. Shrestha, and A. K. Shrivastava, "AI-Driven Conversational Rental Marketplace with Fuzzy Hybrid Search and Trust-Based Moderation," *Proceedings of International Conference on Innovation in Computing, Science, Engineering and Technology*, vol. 3, no. 1, Oct. 2026. doi:10.65091/icicset.v3i1.97

## EDUCATION AND TRAINING

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|---|---|
| **Bachelor of Engineering, Computer Engineering**, Pokhara University (NCIT) | 2023 to 2027 |
| **Node.js Training**, NCIT, Kathmandu | Dec 2024 to Apr 2025 |
| **Senior Secondary (NEB), Science**, Xavier Academy, 77% | 2022 |
| **Secondary (CBSE)**, Modern Indian School, Nepal, 70% | 2020 |